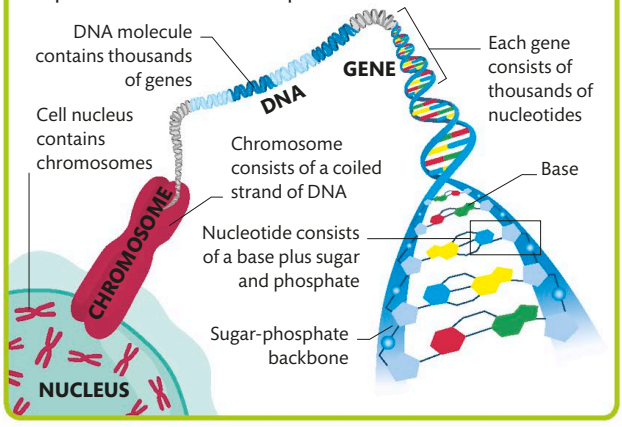
Genetic testing

Genes are segments of DNA—the molecule in our cells that provides the code telling the body how to develop and function. Genetic tests are done to identify any problems that may cause genes to give faulty instructions, including any disorders that may be passed on from parents to children.

HUMAN CELLS
ARE THOUGHT TO
CONTAIN ABOUT **20,000 GENES**

CHROMOSOMES AND GENES

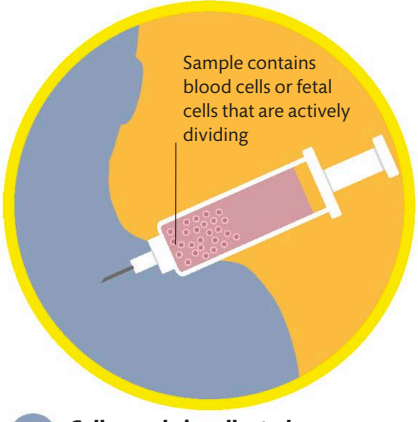
The nucleus of each body cell contains 23 pairs of chromosomes, subdivided into genes. Each gene is made from units called nucleotides. These units have a sugar-phosphate backbone and one of four bases: adenine (A), cytosine (C), guanine (G), or thymine (T). Adenine always pairs with thymine, and cytosine with guanine. The sequence of bases makes up the code of DNA.



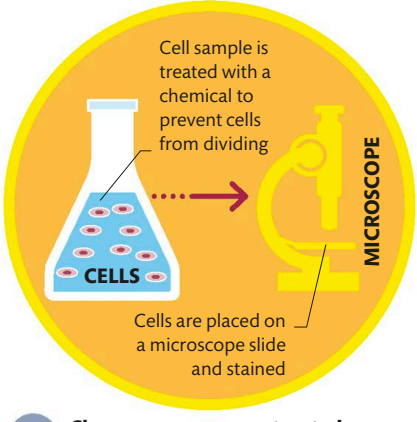
Chromosome testing

Each human body cell has 46 chromosomes—half inherited from the mother and half from the father. Scientists may study the full set of a person's chromosomes, called a karyotype, to see if there are any extra, missing, or abnormal chromosomes.

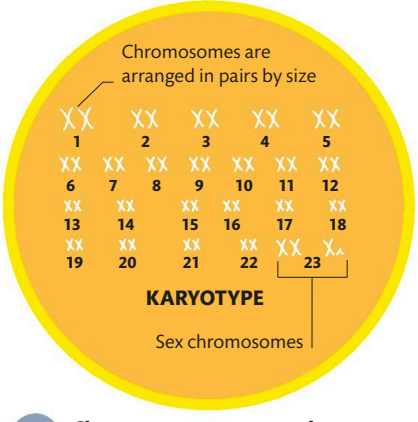
Preparing a karyotype
In karyotyping, chromosomes are studied as cells are dividing to form new cells, when the chromosomes are coiled up into distinctive "X" shapes. The chromosomes are stained, paired, and arranged in order of size to produce the karyotype.



1 Cell sample is collected
A sample of cells is taken from a person's blood or bone marrow. For genetic testing of a fetus, cells are taken from the amniotic fluid or placenta of a pregnant woman.



2 Chromosomes are extracted
The dividing cells are treated with a chemical that prevents them from dividing at a point when the chromosomes are coiled up. The cells are placed on a microscope slide and stained to highlight the chromosomes.



3 Chromosomes are sorted
The chromosomes are sorted and matched into the 22 pairs of autosomes (nonsex chromosomes) and the single pair of sex chromosomes (XX for a female, or XY for a male) to produce the karyotype.